

Decorative Automatic GARDEN WATERING SYSTEM

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The proposed decorative and automatic garden watering device, equipped with a soil sensor module, provides an efficient way to maintain a flourishing garden with minimal human effort while enhancing visual appeal.

The device automates watering by monitoring real-time soil moisture levels. Traditional methods often result in over-watering or under-watering, stressing plants. By employing a soil moisture sensor, the device ensures optimal water

delivery, promoting healthy plant growth and conserving resources.

Fig. 1 illustrates the prototype on a breadboard.

Circuit and working

The circuit diagram for the decorative and automatic garden watering device is shown in Fig. 2. It comprises a step-down transformer (X1), a 12V voltage regulator (LM7812, IC1), a 5V voltage regulator (LM7805, IC2), a soil sensor module with a sensor probe and jumper wire, a PNP transistor (BC558, T1), an NPN transistor (BD139, T2), a 5V SPDT relay, eight different coloured LEDs (LED1 through LED8), and various other components.

The 230V AC mains voltage is stepped down to 15V AC through the step-down transformer (X1). This 15V AC secondary

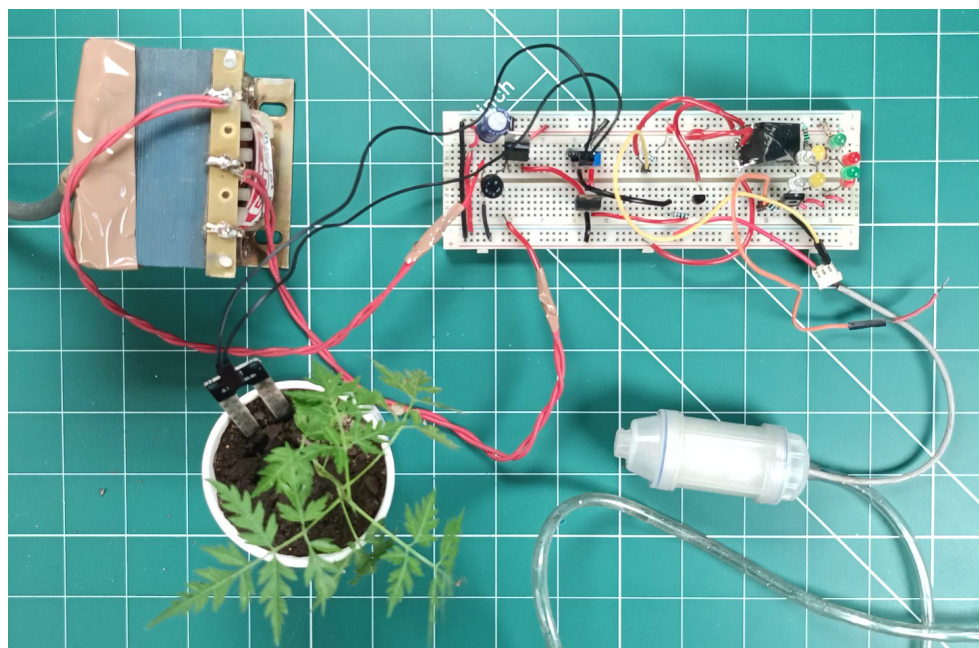


Fig. 1: Author's prototype

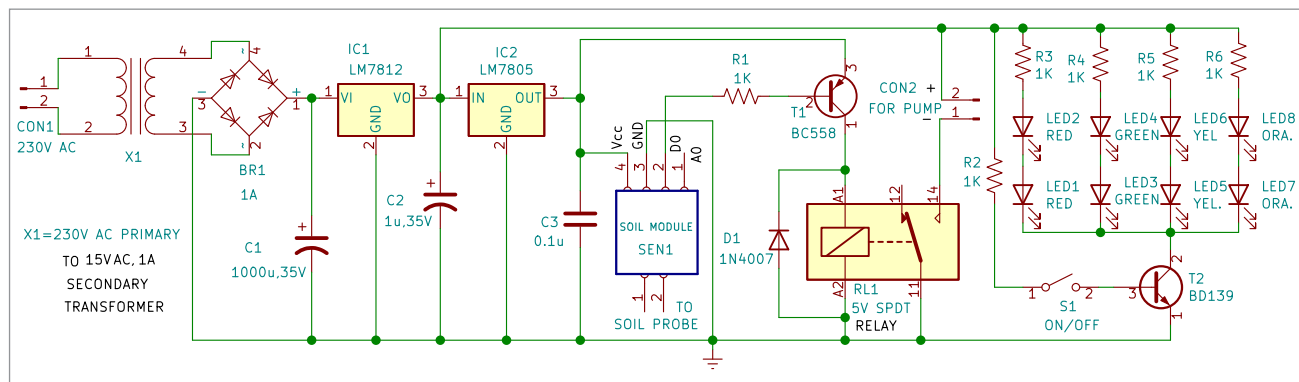


Fig. 2: Circuit diagram